Procedural Lofted Geometry

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1 Introduction

This document gives general information on how to create lofted geometry.

Track is NOT covered in this document. Please see "HOW TO…Create Track" for detailed information on track construction.

A word about the examples in this/these documents.

We used the modeling package 3D Studio max to create Rail Simulator, so this product appears in the examples we have provided. You can, of course, use other packages such as Blender or 3D Canvas. The package we used for authoring textures is Adobe Photoshop, again you may use one of the many other packages available. Regardless of the packages you use, the document should be used as a general guide to the processes to follow to build assets for Rail Simulator

Important note: For your chosen package, you may need additional plug-ins to export the assets into Rail Simulator. We are in the process of finalising these tools and so plug-ins for some packages may not be available immediately.

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2 Overview

2.1 Basics

Procedural lofted geometry covers assets used for:

- Platforms
- Fences
- House-lines
- Tree-lines
- Viaducts

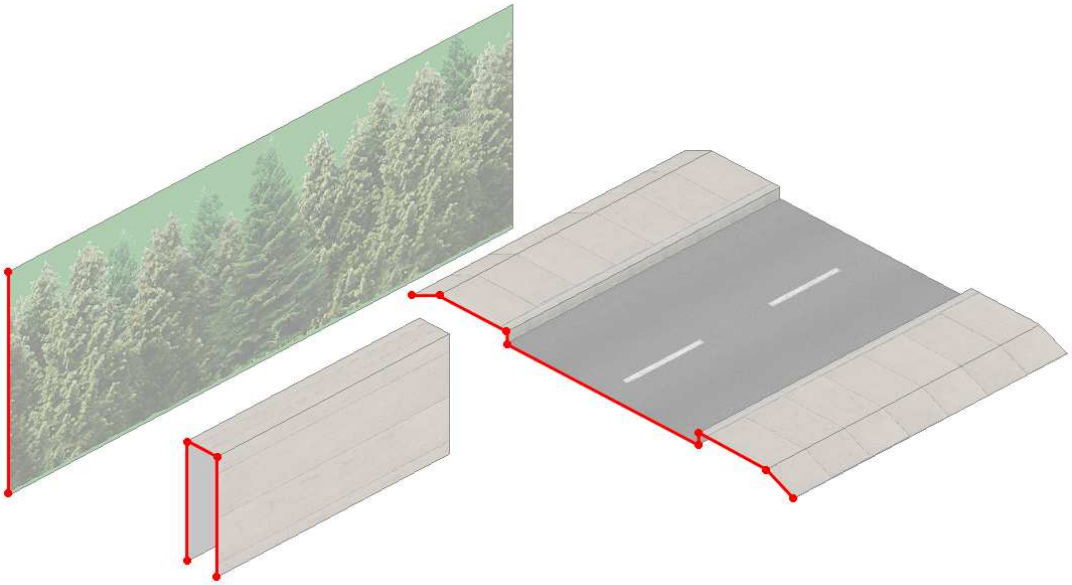
None of these assets can be previewed in the Asset Editor. These assets can only be placed (and hence visible) in the game at runtime.

2.2 Cross-sections

Loft cross-sections are created as extruded splines or edges in Max. These extrusions are exported as IGS files, and then converted as xsec files for use in-game.

There are two important aspects of the extrusion to bear in mind.

- It's important to note that all points of the leading edge of the extrusion (the cross-section) must exist at 0 on the Z plane.
- The cross-section is only extruded in 3DSMax to allow mapping information to be stored (see the section on mapping).



Example Cross-section extrusions

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2.3 Loft Shaders

Lofts must be assigned with specific loft shaders.

There are two loft shaders which can be used. One should be used on solid objects and another should be used on objects which require transparency (see-through areas).

- LoftTexDiff.FX
- LoftTexDiffTrans.FX

NOTE: Remember to set the TRANS flag in the material when using the LoftTexDiffTrans.FX shader.

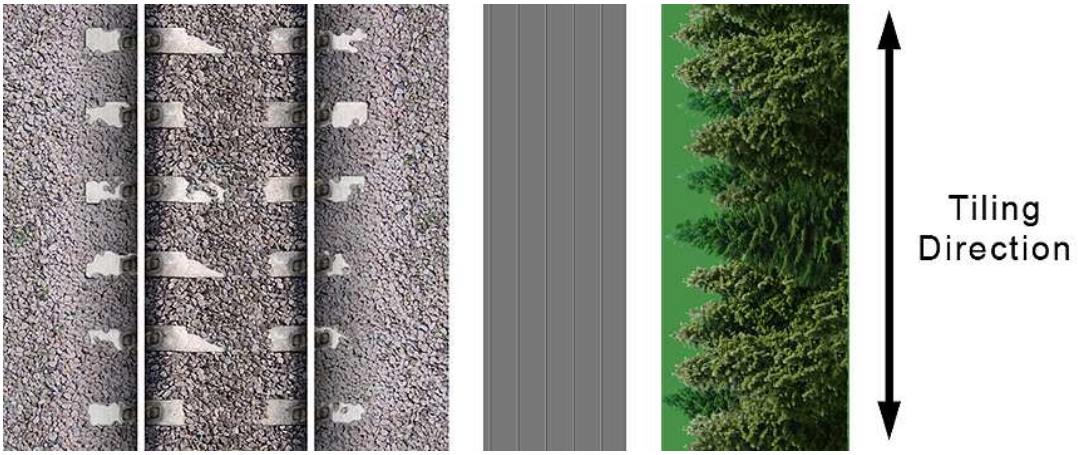
2.4 Mapping

As stated earlier, the cross-section in Max is only extruded to allow mapping information to be exported along with the cross-section data. The length of this extrusion is taken as the distance over which the textures will tile i.e. if the single texture is mapped along the length of the extrusion, and the extrusion is 4m long, then the texture will be tiled once every 4m along the loft when viewed in-game.

To summarise, adjusting the length of the extrusion will compress or stretch the texel size of the in-game loft texture. Generally as a rule, try to keep the texels square.

NOTE: All textures on lofts can only tile in the V direction i.e. when looking at the texture in Photoshop, the texture will only tile in the vertical direction.

On tracks, the rails and ballast do in fact tile in the logical direction (vertically).
However for lofts such as wire fences and distant building-lines the texture needs to be created 90 degrees counter-clockwise to the normal in Photoshop.



Textures showing orientation of 'Rails and Ballast', 'wire fence' and 'treeline

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2.5 Population Geometry

Lofts can have modular pieces of geometry placed at regular intervals along the entire length of the loft. Up to 3 types of geometry can be specified per loft; *start middle and end*.

- The *start* geometry is placed once at the start of the loft.
- The *middle* geometry is placed along the entire length of the loft, populated at a frequency specified in a blueprint.
- The *end* geometry is placed once at the end of the loft.

2.5.1 Examples of uses

Fence posts populated along a fence loft (middle geometry)
Lamp-posts populated along a road loft (middle geometry)
Buffers placed at either end of a section of track (start and end geometry)

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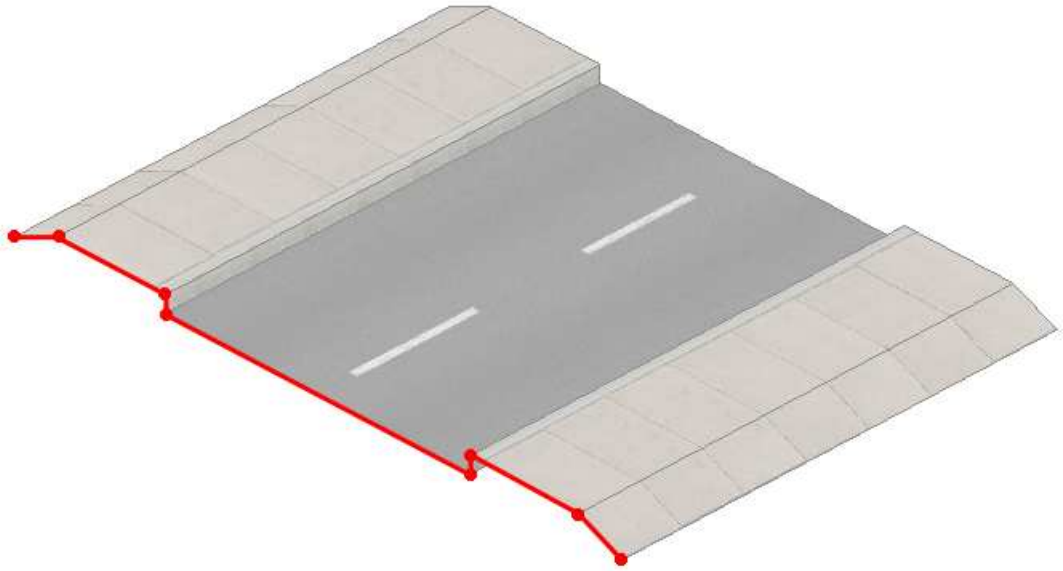
3 Roads

3.1 Road lofts

These assets use *road section blueprints.*

Road lofts are authored and lofted in a similar fashion to the track lofts, but are less complex in structure.



They would generally have a structure as follows:
1_0300_road
2_0800_road
3_1500_road

REMEMBER: There is the potential to place streetlights along the road lofts, as population geometry.

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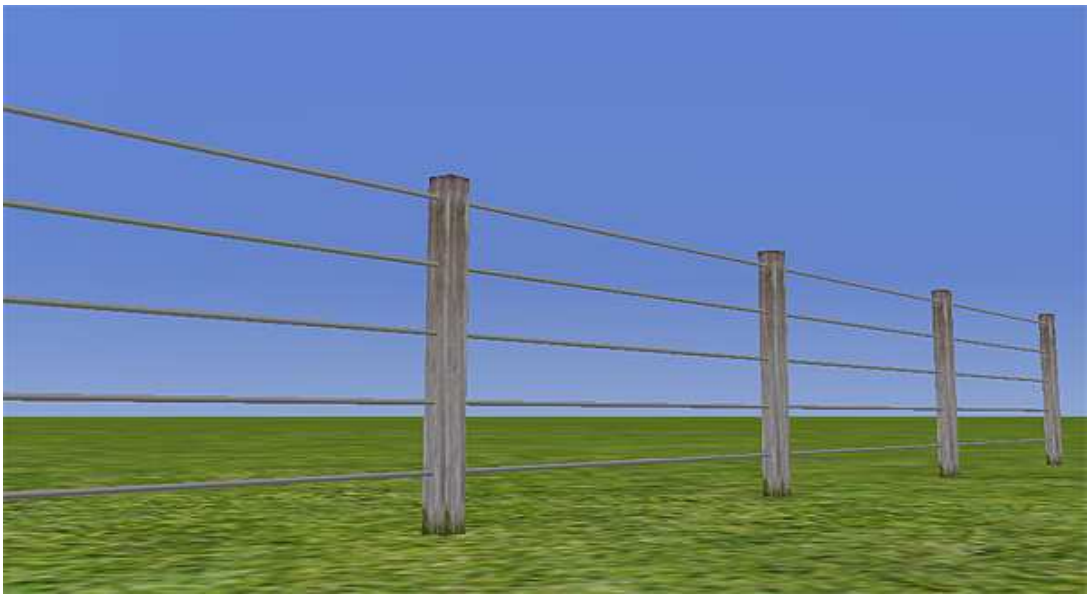
4 Fences

4.1 Basics

These assets use simple *loft section blueprints.*

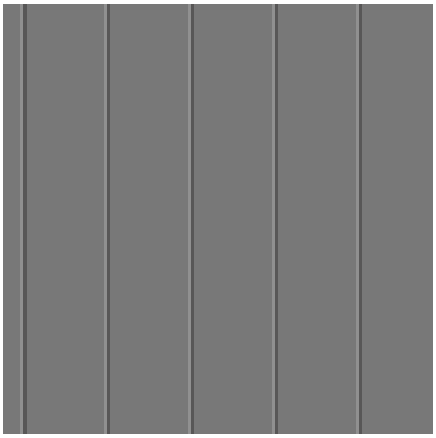
An example fence:



Here we are mapping a single wire fence texture along the loft, using the population geometry to place wooden posts along the loft.

4.2 Texture

Remember to author the texture 90degrees counter-clockwise, and that this loft would use transparency. The texture for the wire fence looks as follows:



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5 Viaducts

Viaducts exist as bespoke or lofted structures.

5.1 Bespoke Structures

These are simply made to measure (usually straight) scenery structures built with a specific location in mind that usually span the gap in one whole section.

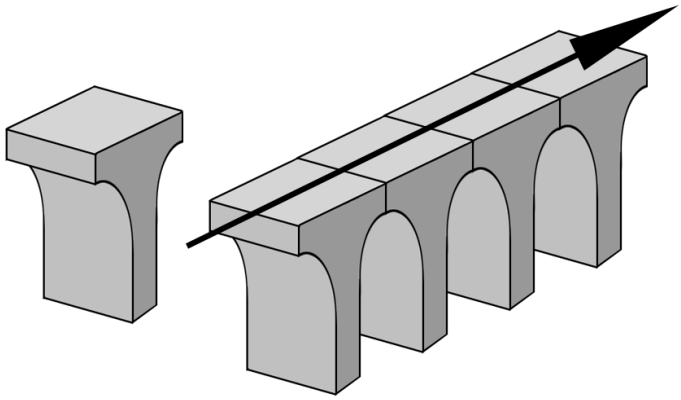
5.2 Lofted Structures

These offer a more general solution where viaducts need to follow a curved path. They can be built as a smaller modular piece (see diagram below) for both a specific location and as a more general structure as they can be made to follow any path using the lofting technology.

5.2.1 How they work

A loft is created using the standard *loft section blueprint*.

The loft itself references a simple cross-section that has a fully transparent texture applied using the LoftTexDiffTrans.FX shader. This will give the correct result of rendering the loft invisible as we don't wish to see the loft itself. We are using this invisible loft to position a series of small modular pieces of a viaduct (see diagram below).



Modular Viaduct piece and the resulting continuous Viaduct structure

5.2.2 Population Geometry

As with other lofts, by specifying the modular viaduct piece asset in the *middle geometry* field in the *loft section blueprint*, this asset will be populated along the loft (which is invisible in this case) at a distance specified as the population frequency.

5.2.3 Population frequency on viaducts

If the viaduct always follows a straight path then the population frequency can be equal to the exact length of the viaduct section. In this case there would be no need for an overlap between each neighbouring section and they would be correctly positioned up against one another.

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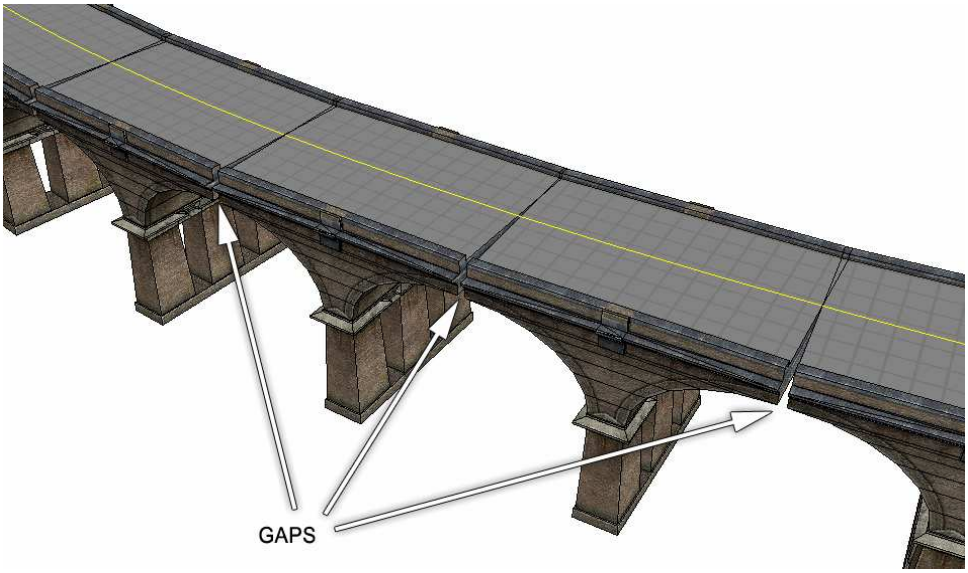
However if the viaduct follows a curved path (as in most cases) and the population frequency is set to the exact length of the viaduct section, then as the viaduct follows the curvature of the loft, small gaps will appear on the outer edge of the curve. See image below.



However a simple solution to this problem is to specify the population frequency as a value which is slightly less than the length of the viaduct section. This will create a small area of overlapping geometry between each neighbouring viaduct section. This overlap area will mask the gap created as the viaduct follows the curvature of the path. See image below.

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